

Supplementary Information: IRSpectra-Bench — protocol, scale tables, cases, and reproducibility

Anonymous authors

Companion to the ICLR 2027 submission

Generation Recall, Not Verification, Binds LLM Structure Elucidation
from Literature Spectra

20 September 2026

This SI holds volume that does not belong on the 9-page ICLR main text. Every accuracy number is copied from a committed lock (docs/HEADLINE_N500_2026-09-20.md, spectro-agent PR #41 48fcd30; docs/POOLED_HEADLINE_2026-09-16.md; docs/NIGHT_POOL_2026-09-20.md). No CIs are invented for $n=500$ / expand-500. No forward-verify precision is claimed on the 106 or the 230. The paper headline is $n=500$ (227/500 top-1; 249/500 recall). Do not quote 243/519. Figure and listing numbers below refer to the main PDF (Fig. 1 = protocol + $n=500$ generation wall; Listing 1 = IReXP payload). They are not re-numbered here.

Contents

1	How to read this SI (figure / listing map)	2
2	Visual plates (reprints; not a second Figure 1)	3
3	Blind protocol and pre-registration	3
3.1	Solver-facing contract	3
3.2	Pre-registration pointers	4
3.3	Stopping / pooling rules	4
3.4	What the solver may do	5
4	Scoring contract (equations used in the main text)	5
5	Former $n=300$ lock and expansion-106	6
5.1	Five flagged +106 compounds	6
5.2	Validate-clean extras (not the headline)	6
6	expand-500 generation (NIGHT_POOL)	7
7	Pool options (generation only)	8
8	Forward-verify coverage (honest gaps)	8
8.1	Locked $n=194$ — complete	8
8.2	Expansion-106 — not run	8
8.3	expand-500 — official score (not a wall)	8

9	Failure and success case panels	9
9.1	Proposal miss — main-R06 / $C_{29}H_{30}FNO_2$	9
9.2	Verify-save — v3-R25 / $C_{10}H_{14}N_2O$	9
9.3	Verify-fail — v3-R26 / $C_{14}H_{11}N_3O_3$	10
9.4	Listing 1 is the IReXP record plate	10
9.5	Extra panels (already-reported numbers)	10
10	Prompt skeletons	11
10.1	Headline solver-facing skeleton	11
10.2	Forward-verify skeleton	12
10.3	Cross-vendor harness	12
10.4	Deposit schema	12
11	Hyperparameters, hardware, and model cards	13
11.1	Headline model card	13
11.2	Cross-vendor card ($n=60$)	13
11.3	Literature decomposition (copied, not re-derived)	13
12	Two-paper split and anonymity	14
13	Reproducibility checklist	14
13.1	What a reviewer can re-check without the withheld keys	15
14	Reviewer FAQ (SI copy)	15

1 How to read this SI (figure / listing map)

The night main-text plate and listing are the source of truth. This SI does not fight that numbering.

Table 1: Main-text floats vs this SI. Fig. 1 is the Figma Bro lead plate.

main label	object	SI role
Fig. fig1	Fig. 1 (fig1_lead_overview)	Figma Bro lead: input $\rightarrow$ generate $\rightarrow$ verify, factorisation, and $n=500$ generation wall 227/22/251.
Fig. fig-chemspace	Fig. 2	$n=300$ chemistry; panels on validate-clean $n=295$.
Listing lst:payload	Listing 1 (listing1_irexp_record)	IReXP JSON+molecule (NTJIYHYWVZY-BEX; $C_{12}H_{16}N_2O_2$). SMILES/InChIKey are resource fields, not solver payload.
Fig. fig-wall	appendix (fig_wall_diagnostic)	Repeat of the Fig. 1 $n=500$ generation wall (227/22/251).
Fig. fig-robustness	Fig. 4	Formula-only / recency / four-vendor.
Fig. fig-mechanism	Fig. 5	Locked v3-R25 verify-save. Reused in §9.
Fig. fig3-method	Fig. 6	Inference ladder on the 60-compound arm.
App. Tables 7–10	cases / +106 / $n=295$ / scale	Same integers as §5–7.
SI Figs. S1–S3	reprints	Same three files as main Fig. 1 / Listing 1 / appendix wall (§2). Not a second lead figure.

Equations in the main text are Eqs. (1)–(7): factorisation, InChIKey-14 indicator, top-1, recall@ k , prec.|recall, ^{13}C chamfer, pooled estimator. They are restated in §4 with the same meaning. No new metric is defined.

2 Visual plates (reprints; not a second Figure 1)

The files below are the same as the main PDF. They are reprinted so this SI is readable alone. Main-text Figure 1 (fig1_lead_overview) remains the only lead diagram. Panel letters (a)/(b)/(c) from an earlier draft do not apply: the Figma Bro plate is one figure (protocol strip, factorisation, wall).

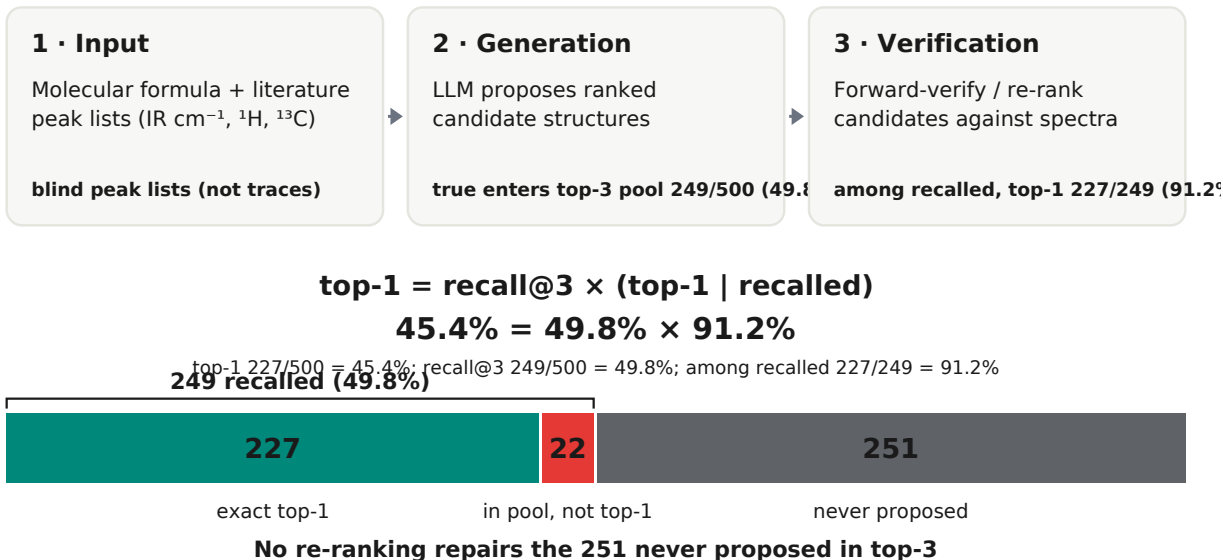


Figure S1: Reprint of main-text Figure 1 (fig1_lead_overview). Input → generate → verify; 45.4% = 49.8% × 91.2%; $n=500$ generation wall 227/22/251 (249 recalled, 49.8%).

3 Blind protocol and pre-registration

3.1 Solver-facing contract

Each IRSpectra-Bench item is a four-field JSON object. The solver sees only

$\{\text{qid}, \text{formula}, \text{ir_bands_cm-1}, \text{h_nmr}, \text{c_nmr}\}$.

No name, SMILES, InChIKey, scaffold, atom-type hint, retrieved spectrum, licence stamp, or PMC id enters the prompt (main-text Listing 1, adapted from one IReXP record). Gold constitution is scored offline by the InChIKey-14 indicator (§4).

Difficulty is declared a priori by RDKit ring analysis: simple iff ≤ 2 rings, no fused/spiro/bridge-head system, ≤ 22 heavy atoms; else complex. The $n=500$ headline is near-50/50 (248/252); former

Listing 1. Example IReXp JSON record (commercial DoR).

```
{
  "id": "NTJIYHYWVZYBEX-UHFFFAOYSA-N",
  "ir_bands_cm-1": [3337.0, 2969.0, 1633.0, 1556.0,
    1505.0, 1240.0, 1181.0, 1071.0, 826.0],
  "ir_source": "experimental",
  "source_doi": "PMC:13029360",
  "pmcid": "PMC13029360",
  "h_nmr": "8.07 (brs, 1H, NH), 7.25 (d, J = 7.3 Hz, 2H, Ar), 6.78 (d,
    J = 7.3 Hz, 2H, Ar), 6.25 (d, J = 8.1 Hz, 1H, NH), 4.10
    (hex, J = 8.1 Hz, 1H, CH), 3.66 (s, 3H, OCH3), 2.23-2.09
    (m, 2H, CH2), 1.90-1.71 (m, 2H, CH2), 1.64-1.51 (m, 2H,
    CH2)",
  "c_nmr": "154.8, 154.4, 134.0, 119.9, 114.3, 55.5, 45.0, 31.8, 14.8",
  "smiles": "C0c1ccc(NC(=O)NC2CCC2)cc1",
  "inchikey": "NTJIYHYWVZYBEX-UHFFFAOYSA-N",
  "has_structure": true,
  "license": "CC-BY",
  "license_pool": "commercial"
}
```

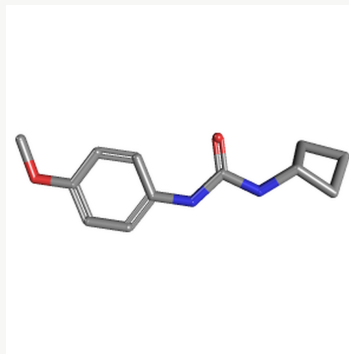


Figure S2: Reprint of main-text Listing 1 (listing1_irexp_record). IReXp JSON+molecule ($C_{12}H_{16}N_2O_2$; NTJIYHYWVZYBEX-UHFFFAOYSA-N). SMILES / InChIKey are resource fields, not solver payload. Not a Data Descriptor dump. Locked v3-R25 is the mechanism case (Figure S4), not this listing.

$n=300$ was 151/149. the eligible IReXp corpus is 17.5% simple / 82.5% complex. Reweighting lives on validate-clean $n=295$ (§5) and is not an $n=500$ figure.

3.2 Pre-registration pointers

These files live in the companion spectro-agent tree, not this postcard manuscript repo. Pointers are recorded so a reviewer can ask for the committed note rather than a reconstructed story.

- Locked $n=194$: main clean 134 + v3 40 + v2_ctrl 20. Spectrally validated main round plus two control arms. Sampler seeds for the original 140 / 40 draws were not captured (gap #4 in docs/SUBMISSION.md); the released roster is fixed and reusable but not exactly re-drawable.
- Expansion-106: pre-registered draw after the 194 lock, same irexp_resolved sampler. Five ^{13}C -overread flags R12, R22, R25, R82, R91 stay in the $n=500$ headline (and in former $n=300$). Validate-clean $n=295$ is appendix / SI sensitivity.
- expand-500: docs/EXPANSION_PREREGISTRATION_500.md (seed 2026500; $n=230$; 115/115 simple/complex; exclusion of 375 prior InChIKey-14s including a local restore of the withheld +106 key). 0 collisions with any prior round. Six ^{13}C -overread flags known before any solve: R26, R31, R102, R105, R107, R138.

3.3 Stopping / pooling rules

- The $n=500$ 200-cut is the pre-specified clean-first string-sorted prefix (R01–R75; docs/headline500_expand200_qids.json), not an outcome-selected subset and not all 230.
- score_pooled.py –expand-500 refuses a missing predictions2.jsonl or a withheld key.
- answers2.jsonl is not in either working tree (git ls-files empty for +106 and expand-500).

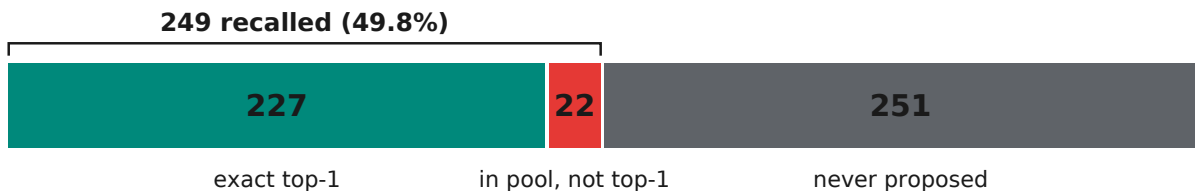


Figure S3: Reprint of the appendix diagnostic wall (fig_wall_diagnostic): 227 exact top-1 / 22 in pool, not top-1 / 251 never proposed on $n=500$. Bracket: 249 recalled (49.8%). Same integers as Figure S1.

- Keys were restored only under /tmp/blind/_key/ for the 2026-09-20 score pass, then deleted.
- validate_benchmark.py was not re-run on the night pass (would rewrite committed clean_qids.json). The existing 224-qid snapshot is used.
- collect_round.py was not run; 230 raw/single_R*.json files were left untouched.
- forward_verify_main.py score did run on expand-500 (§8): 103/230 verify vs 129/230 self. Official prep was not re-run.
- -ci / bootstrap was not run on expand-500. Do not invent CIs.

3.4 What the solver may do

RDKit formula / parse check only. No web search. No retrieved spectrum. No thinking tier on the locked 194 and +106 arms. The expand-500 200-cut used a thinking-tier consumer arm and is labelled as such — it is in the $n=500$ mix and is not interchangeable with the no-thinking 194+106 protocol.

A deposit is JSONL of {qid, candidates: [SMILES,...]} under a stated budget. Headline budget: ≤ 3 ranked SMILES per qid. Self-ranking vs forward-verified re-ranking must be labelled; they are not interchangeable.

4 Scoring contract (equations used in the main text)

These are the same identities as main-text Eqs. (1)–(7). They are restated so the SI is readable alone. No new metric is introduced.

A candidate $\hat{y}$ matches gold y^* at constitution iff

$$s(\hat{y}, y^*) = \mathbf{1}[\text{IK}_{14}(\hat{y}) = \text{IK}_{14}(y^*)]. \quad (1)$$

IK_{14} is the first 14 characters of the RDKit InChIKey (connectivity layer). Full stereo-sensitive InChIKey is reported only on validate-clean $n=295$ (§5).

With ranked pool $\mathcal{C}_i = (c_{i1}, \dots, c_{ik})$, $k \leq 3$,

$$\text{top-1} = \frac{1}{n} \sum_i s(c_{i1}, y_i^*), \quad (2)$$

$$\text{recall}@k = \frac{1}{n} \sum_i \mathbf{1}[\exists j \leq k : s(c_{ij}, y_i^*) = 1], \quad (3)$$

$$\text{prec.}|\text{recall} = \frac{\sum_i s(c_{i1}, y_i^*)}{\sum_i \mathbf{1}[y_i^* \in \mathcal{C}_i]}. \quad (4)$$

These are the three reportable numbers. They factorise as

$$\text{top-1} = (\text{recall}@k) \times (\text{prec.}|\text{recall}) \quad (5)$$

when the same pool is used for both stages — the rates still have different denominators and must not be differenced. Self-ranking prec.|recall on $n=500$ is 227/249 (91.2%). That is not forward-verify precision.

Forward-verify re-ranks by symmetric chamfer on ^{13}C peak sets $P, Q \subset \mathbb{R}$

$$d(P, Q) = \frac{1}{|P|} \sum_{p \in P} \min_{q \in Q} |p - q| + \frac{1}{|Q|} \sum_{q \in Q} \min_{p \in P} |q - p|, \quad (6)$$

where $\hat{P}(c)$ is predicted blind to the observed list P_{obs} . Instrumented fverify only on the locked $n=194$ (58/65). Expansion-106: not run. expand-500 official score is in §8 (103/230 verify vs 129/230 self; no CIs; not a wall).

A pool estimator on an index set $\mathcal{S}$ is

$$\hat{p}_{\mathcal{S}} = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} s(c_{i1}, y_i^*). \quad (7)$$

The ICLR headline uses $\mathcal{S}$ of size 500 (194 locked + all 106 Opus + 200 expand-500; R01–R75 cut). Sensitivity $\mathcal{S}$ of size 300 / 524 / 530 (§7) do not replace that headline. Do not quote the $n=519$ score_pooled.py print (243/519) as the headline — that pool drops five +106 flags.

Confidence intervals on former $n=300$ / $n=295$ / $n=194$ are bootstrap 95% (score_main.boot, seed 0, 2000 resamples). Paired model differences use McNemar with Holm correction. No CIs are reported for $n=500$, expand-500, or the sensitivity pools.

5 Former $n=300$ lock and expansion-106

Source: docs/POOLED_HEADLINE_2026-09-16.md (python scripts/score_pooled.py on spectro-agent). These rows are the former ICLR lock; the paper headline is now $n=500$ (§7).

Self-ranking precision | recall on the generation pool: 118/133 (88.7%) on former $n=300$; 116/130 (89.2%) on $n=295$; 227/249 (91.2%) on $n=500$ — not forward-verify precision.

5.1 Five flagged +106 compounds

5.2 Validate-clean extras (not the headline)

These were scored on $n=295$ only. They were not re-run on the $n=300$ roster and must not be quoted as $n=300$ figures.

On the locked 194, of 137 analysable top-1 misses, 76.6% are constitutional isomers of the true structure; only 22.6% share the true Murcko scaffold. Named-ring annotation on that slice: 10/194 author assignments name a ring system, solved 1/10 vs 54/184 for the rest. Neither analysis was re-run on the expansion. Fable expansion: 68/106 deposits; not scored; never pooled. Expansion-106 fverify: not run.

Table 2: Former $n=300$ generation slices (now appendix).

set	n	top-1	recall@3	notes
locked 194	194	55/194 (28.4%) [22–35]	65/194 (33.5%) [27–40]	instrumented fverify
locked simple	98	47/98 (48.0%)	53/98 (54%)	
locked complex	96	8/96 (8.3%)	12/96 (12%)	
expansion-106 all	106	63/106 (59.4%) [50–69]	68/106 (64.2%) [56–73]	enters headline
+106 simple	53	41/53 (77.4%)	—	
+106 complex	53	22/53 (41.5%)	—	
expansion-106 clean	101	61/101 (60.4%) [50–69]	65/101 (64.4%) [54–73]	appendix $n=295$
clean simple	49	40/49 (81.6%)	41/49 (84%)	
clean complex	52	21/52 (40.4%)	24/52 (46%)	
former 194+all 106	300	118/300 (39.3%) [34–45]	133/300 (44.3%) [39–50]	now appendix
former simple	151	88/151 (58.3%)	96/151 (64%)	
former complex	149	30/149 (20.1%)	37/149 (25%)	
validate-clean 194+101	295	116/295 (39.3%) [34–45]	130/295 (44.1%) [39–49]	appendix
clean-pool simple	147	87/147 (59.2%) [51–67]	94/147 (63.9%) [56–71]	
clean-pool complex	148	29/148 (19.6%) [14–26]	36/148 (24.3%) [17–31]	

Table 3: +106 ^{13}C -overread flags. Included in $n=300$; out of $n=295$.

item	value
qids	R12, R22, R25, R82, R91
reason	^{13}C -overread (scripts/validate_benchmark.py)
snapshot	data/benchmark_expand/clean_qids.json
when flagged	before subset totals (stopping rule)
in paper headline?	yes (in $n=500$ and former $n=300$)
effect vs excluding	+2 top-1, +3 recall $\rightarrow$ 118/300 and 133/300
one-decimal top-1	unchanged (both 39.3%)

6 expand-500 generation (NIGHT_POOL)

Source: spectro-agent data/benchmark_expand_500/NIGHT_POOL_2026-09-20.md on cursor/expand-bench-500-b78b (copied to docs/NIGHT_POOL_2026-09-20.md). Key restored only for score2, then deleted. answers2.jsonl is not in the tree.

Mean best Tanimoto (score2 footer, all 230): simple 0.863; complex 0.645. Flagged qids (in 230, out of 224): R26, R31, R102, R105, R107, R138. Deposits: 230/230; predictions2.jsonl 230 lines, 690 candidates. Unique (formula, IR, ^{13}C) vs irexp_resolved.jsonl.gz: 230/230; 0 prior-round InChIKey-14 collisions. Protocol: thinking-tier consumer arm — not interchangeable with the no-thinking 194+106 protocol.

Verbatim score2 footer (do not polish the integers):

overall recovered (top-3): 138/230 (60%) top-1 exact: 129/230 (56%)
 simple : recovered 87/115 (76%) top1 82/115 (71%) meanBestTani 0.863
 complex : recovered 51/115 (44%) top1 47/115 (41%) meanBestTani 0.645

Table 4: Validate-clean $n=295$ extras. Same scorer; not the headline.

metric		overall	simple	complex
scaffold-level (best Tanimoto ≥ 0.45)		64%	80%	48%
mean best Tanimoto		0.66	0.79	0.53
full InChIKey (stereo) top-1	93/295 (31.5%) [26–37]		—	—
full InChIKey recall	108/295 (36.6%) [31–42]		—	—
corpus-reweighted top-1	26.5% [21–32]		—	—
corpus-reweighted recall	31.3% [25–37]		—	—
locked-194 stereo top-1	41/194 (21.1%)		—	—

Table 5: Size bins on validate-clean $n=295$.

heavy atoms	n	top-1	recall
≤ 15	53	36/53 (67.9%)	39/53 (73.6%)
16–25	152	65/152 (42.8%)	75/152 (49.3%)
> 25	90	15/90 (16.7%)	16/90 (17.8%)

7 Pool options (generation only)

The estimator is Eq. 7. Only the first row is the paper headline. Source: docs/HEADLINE_N500_2026-09-20.md (spectro-agent 48fcd30). 200-cut list: docs/headline500_expand200_qids.json (R01–R75).

Strata on the $n=500$ headline (same InChIKey-14 rule): simple 161/248 (65%) top-1 / 172/248 (69%) recall; complex 66/252 (26%) / 77/252 (31%). 200-alone: 73/97 and 76/97 simple; 36/103 and 40/103 complex.

Do not treat 524 or 530 as a replacement headline. Do not quote score_pooled.py –expand –expand-500 ($n=519$, 243/519) — that path clean-filters both expansions and drops five +106 flags. Do not draw a pooled wall from these rows. Do not attach CIs.

8 Forward-verify coverage (honest gaps)

8.1 Locked $n=194$ — complete

Source: data/diagnosis.json / scripts/forward_verify_all.py.

This 58/7/129 triple is the instrumented $n=194$ protocol-slice diagnostic. It is not the main-text Figure 1 / appendix wall (those are the $n=500$ generation decomposition 227/22/251; Figure S1, Figure S3).

8.2 Expansion-106 — not run

No data/fverify_expand/ bundle. No precision number. Self-ranking 227/249 on $n=500$ is not a substitute.

8.3 expand-500 — official score (not a wall)

Source: spectro-agent data/fverify_expand_500/STATUS.md, results.txt, and NIGHT_POOL_2026-09-20.md (cursor/expand-bench-500-b78b). Blind ^{13}C campaign over 681 unique candidate SMILES

Table 6: expand-500 generation (score2, InChIKey-14). No CIs.

set	<i>n</i>	top-1	recall@3
all deposited Opus	230	129/230 (56.1%)	138/230 (60.0%)
simple	115	82/115 (71%)	87/115 (76%)
complex	115	47/115 (41%)	51/115 (44%)
validate-clean	224	127/224 (56.7%)	135/224 (60.3%)
simple	110	81/110 (74%)	85/110 (77%)
complex	114	46/114 (40%)	50/114 (44%)

Table 7: Pool options. Headline is the first row. No CIs. No fverify.

<i>n</i>	composition	role	top-1	recall
500	194+all 106+200 (R01–R75)	paper headline	227/500 (45.4%)	249/500 (49.8%)
200-cut	clean-first string-sorted expand-500	in the headline	109/200 (54.5%)	116/200 (58.0%)
300	194+all 106	former headline (appendix)	118/300 (39.3%)	133/300 (44.3%)
524	194+all 106+clean 224	sensitivity	245/524 (46.8%)	268/524 (51.1%)
530	194+all 106+all 230	sensitivity (do not use 230 as 500)	247/530 (46.6%)	271/530 (51.1%)
295	194+clean 101	older lock; appendix	116/295 (39.3%)	130/295 (44.1%)

(f1–f41 all present). Official forward_verify_main.py score() ran 2026-09-20 (is_true filled under /tmp only; committed candidates.jsonl still omits it; official prep was not re-run). No CIs. Verify does not beat self-rank. Headline is $n=500$ generation. This is not a pooled wall and does not replace the $n=194$ diagnosis.json plate.

Coverage: 41/41 batches; 681/681 unique SMILES; 230/230 qids fully covered. Multi-candidate only: all-230 self 128/137, verify 102/137; clean-224 self 126/134, verify 102/134. forward_verify_all.py was not run (no expand-500 diagnosis.json).

HOSE-style lookup on the locked $n=194$ candidate sets ties self-ranking; a modest GNN reaches 59/65 (91%) vs the LLM verifier’s 58/65 (89%) — directional, not a new headline. Deranging observed ^{13}C pairings collapses precision ($p=0.001$).

9 Failure and success case panels

The three locked-slice modes in the main appendix, plus extra panels that use only already-reported numbers. IDs below are locked-round labels (expansion qids reuse R22/R25). No SMILES are invented for R06/R26; only v3-R25 molecules are drawn (Figure S4).

9.1 Proposal miss — main-R06 / $\text{C}_{29}\text{H}_{30}\text{FNO}_2$

True acridinedione never enters the 3-candidate pool (Tanimoto-0.90 regioisomer: 4-fluorophenyl / phenyl swapped). IR 1641, 1510 cm^{-1} ; ^{13}C 195.7. Verifier unused. Never-proposed bind on the $n=194$ fverify slice: re-ranking cannot recover a structure that was never proposed (Eq. 5).

9.2 Verify-save — v3-R25 / $\text{C}_{10}\text{H}_{14}\text{N}_2\text{O}$

True 2-pyridyl picolinamide at self-rank 3; self-rank 1 is 3-pyridyl. Forward-verify recovers (chamfer 0.42 vs 1.30 ppm, Eq. 6). IR 1679 cm^{-1} . Main-text Figure 5; SI Figure S4.

Table 8: Sensitivity-pool strata. Copied from HEADLINE_n500 / NIGHT_POOL.

pool	simple n	simple top-1 / recall	complex n	complex top-1 / recall
500	248	161/248 (65%) / 172/248 (69%)	252	66/252 (26%) / 77/252 (31%)
300	151	88/151 (58%) / 96/151 (64%)	149	30/149 (20%) / 37/149 (25%)
524	261	169/261 (65%) / 181/261 (69%)	263	76/263 (29%) / 87/263 (33%)
530	266	170/266 (64%) / 183/266 (69%)	264	77/264 (29%) / 88/264 (33%)

Table 9: Instrumented forward-verify protocol-slice on $n=194$ (diagnosis.json). This 58/7/129 triple is not the main-text wall. expand-500 has an official score in the next subsection, not a wall.

	60-compound arm	locked slice ($n=194$)
generation recall	19/60 (32%)	65/194 (34%)
top-1, solver self-ranking	14/60 (23%)	55/194 (28%)
top-1, forward-verified re-ranking	16/60 (27%)	58/194 (30%)
precision recall — fverify	16/19 (84%)	58/65 (89%)
multi-candidate only — fverify	10/13 (77%)	30/37 (81%)
verified / mis-ranked / never proposed	—	58 / 7 / 129
derangement floor on $n=37$	—	54.0%; fverify +27.1 points
McNemar vs self-rank	—	$p=0.55$ (7 gained, 4 lost)

9.3 Verify-fail — v3-R26 / C₁₄H₁₁N₃O₃

True 3-nitrophenyl dihydroquinazolinone in pool (self-rank 2); 2-nitro ranked first. Chamfer 1.35 vs 1.36 ppm. IR 1652, 1530, 1352 cm⁻¹. Ranking can fail when predicted ¹³C lists are nearly tied. This is why we do not claim an accuracy advance from forward-verify (McNemar $p=0.55$; 7 gained, 4 lost on $n=194$).

9.4 Listing 1 is the IReXP record plate

Main-text Listing 1 is listing1_irexp_record (C₁₂H₁₆N₂O₂ urea; InChIKey NTJIYHYWVZYBEX-UHFFFAOYSA-N). SMILES / InChIKey / licence fields are resource metadata. Bench solvers see only formula and printed peak lists. Gold constitution is scored only by Eq. 1. Locked v3-R25 remains the mechanism worked case (main-text Figure 5), not Listing 1.

The plate is reprinted as Figure S2.

9.5 Extra panels (already-reported numbers)

Formula-only collapse (control, $n=60$). Formula only: 3/60 (5%) top-1 and 3/60 (5%) recall. Formula + IR + ¹H + ¹³C: 14/60 (23%) / 19/60 (32%). Outcomes perfectly nested (McNemar $p=0.001$). The model is reading the peak lists, not only the formula.

Generate-wide (60-compound arm). Ten solver agents, up to six candidates each, pooled then re-ranked: recall 32% → **42%**, top-1 23% → **30%** (McNemar $p=0.34$); verification precision falls 84% → 72%. The training-free ceiling remains recall-limited. This is a success-for-recall panel, not an accuracy claim.

Table 10: expand-500 official chamfer score. Copied from NIGHT_POOL / fverify STATUS. Not a wall. Not the headline.

set	<i>n</i>	recall	self top-1	verify top-1	verify recall
all deposited	230	138/230 (60.0%)	129/230 (56.1%)	103/230 (44.8%)	103/138 (74.6%)
validate-clean	224	135/224 (60.3%)	127/224 (56.7%)	103/224 (46.0%)	103/135 (76.3%)

Table 11: Three-mode case grid (disclosed facts only). Same integers as main-text Table tab:cases.

mode	id / formula	disclosed facts
Proposal miss	main-R06 / C ₂₉ H ₃₀ FNO ₂	True acridinedione never in the 3-candidate pool (Tanimoto-0.90 regioisomer: 4-fluorophenyl / phenyl swapped). IR 1641, 1510 cm ⁻¹ ; ¹³ C 195.7. Verifier unused. Never-proposed bind on the <i>n</i> =194 fverify slice.
Verify-save	v3-R25 / C ₁₀ H ₁₄ N ₂ O	True 2-pyridyl picolinamide at self-rank 3; self-rank 1 is 3-pyridyl. Forward-verify recovers (chamfer 0.42 vs 1.30 ppm). IR 1679 cm ⁻¹ . Figure S4. Not Listing 1.
Verify-fail	v3-R26 / C ₁₄ H ₁₁ N ₃ O ₃	True 3-nitrophenyl dihydroquinazolinone in pool (self-rank 2); 2-nitro ranked first. Chamfer 1.35 vs 1.36 ppm. IR 1652, 1530, 1352 cm ⁻¹ .

IRexp-fine-tuned generator (locked slice). Pooled with Claude: recall 33.5% → 54.1%, top-1 28.4% → 35.1% (McNemar *p*=0.015). The wall is a property of training-free elicitation, not of the task. Training is the one intervention already seen to move $P(\text{true} \in \text{pool})$.

Battery-electrolyte subset. *n*=46 literature electrolyte chemistry (not operando): 26% top-1, recall-bound — same regime as the locked slice.

Four-model Claude ladder (*n*=24). Haiku ⊂ Sonnet ⊂ Opus ⊂ Fable is capability-sensitive but underpowered for adjacent ranks, with a disclosed protocol asymmetry. Not a headline.

Expansion difficulty gap. +106 is easier for this solver than the locked 194 (59.4% vs 28.4% top-1). That gap is a result, not a reason to hide the 106 or to keep *n*=194 as the headline. Pooling to *n*=500 does not erase it (main-text Table 2).

10 Prompt skeletons

10.1 Headline solver-facing skeleton

Wrapping instruction text was not captured on the headline Claude arm (Limitations (i); docs/-SUBMISSION.md gap #1). Only the per-compound four-field payload is released. The skeleton below is the contract, not a recovered verbatim system prompt.

You are given only:

formula (as from HRMS)

ir_bands_cm-1 (author-transcribed experimental list)

h_nmr, c_nmr (printed shift lists)

Return <= 3 ranked SMILES.

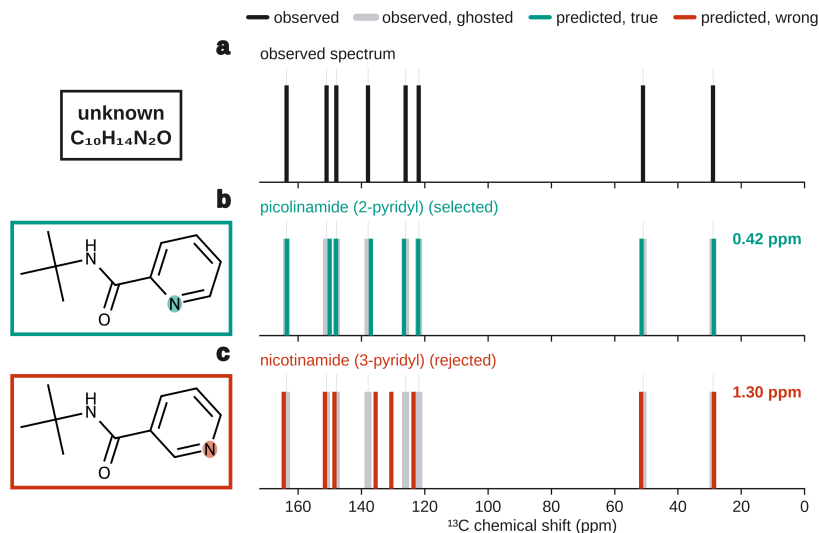


Figure S4: Locked v3-R25 verify-save (main-text Figure 5). Self-rank prefers 3-pyridyl; forward ^{13}C recovers 2-pyridyl (chamfer 0.42 vs 1.30 ppm). A verify-save, not a recall win. Not Listing 1.

No name, no SMILES hint, no scaffold, no retrieved spectrum.
Check molecular formula with RDKit before emitting a candidate.

Listing 1 payload (fields only; ^1H abbreviated with ellipsis in the main float; gold held out):

```
{
  "formula": "C12H16N2O2",
  "ir_bands_cm-1": [3337.0, 2969.0, 1633.0, 1556.0,
    1505.0, 1240.0, 1181.0, 1071.0, 826.0],
  "h_nmr": "8.07 (brs, 1H, NH), 7.25 (d, J = 7.3 Hz, 2H, Ar), ...",
  "c_nmr": "154.8, 154.4, 134.0, 119.9, 114.3, 55.5, 45.0, 31.8, 14.8"
}
```

10.2 Forward-verify skeleton

For each candidate SMILES, predict a ^{13}C peak list without seeing the observed list. Re-rank by $d(\hat{P}(c), P_{\text{obs}})$ (Eq. 6). The instruction dispatched to the forward-prediction sub-agents was likewise not captured; only the candidate SMILES and the resulting numeric lists are on disk for $n=194$.

10.3 Cross-vendor harness

The verbatim prompts in the (unreleased-here) ESI S2 of the spectro-agent tree are the cross-vendor harness prompts, not the missing headline Claude wrapper. External submissions must document model version, prompt, and tool access (docs/LEADERBOARD.md).

10.4 Deposit schema

```
{"qid": "R01", "candidates": ["SMILES_rank1", "SMILES_rank2", "SMILES_rank3"]}
```

Score locally (spectro-agent):

```
python scripts/score_submission.py --predictions my_run.jsonl --name "YourModel-1.0"
python scripts/score_pooled.py          # former n=300 = 194+all 106; not n=500
```

```
python scripts/score_pooled.py --stereo # 93/295 on validate-clean
python scripts/score_main.py # locked n=194
python scripts/forward_verify_all.py # n=194 instrumented slice
```

11 Hyperparameters, hardware, and model cards

11.1 Headline model card

Table 12: Headline solver. Gaps are recorded, not filled.

item	value
headline solver	Claude Opus, consumer subscription
thinking tier (194+106)	none; 200-cut is thinking-tier
snapshot / seed / temperature / top_p / top_k / max_tokens	not exposed; not recorded (SUBMISSION gaps #2-#3)
candidate budget	≤ 3 SMILES / qid
tool access	RDKit formula / parse check only; no web search
fine-tuning	none on the headline arm
paid API	no, for the core protocol
hardware	consumer harness; no cluster job for inference
scoring hardware	commodity CPU; RDKit InChIKey-14
cross-vendor arm	$n=60$; Grok 4.6, Gemini 3.7 Flash, GPT-5.6 Sol
Claude mean candidates on that arm	2.20 (others 3.00) — recall rankings approximate
expand-500 solver	Claude Opus thinking-tier (declared; not headline)
Fable +106	68/106 deposits; not scored; not pooled
scoring scripts	scripts/score_pooled.py, score_main.py, bench-mark_v2.py score2

11.2 Cross-vendor card ($n=60$)

Table 13: Cross-vendor decomposition. Recall and precision use different denominators.

model	generation recall	prec./recall	multi-candidate only
Claude Opus	19/60 = 32%	16/19 = 84%	10/13 = 77%
Grok 4.6	32/60 = 53%	20/32 = 62%	20/32 = 62%
Gemini 3.7 Flash	30/60 = 50%	22/30 = 73%	22/30 = 73%
GPT-5.6 Sol	25/60 = 42%	17/25 = 68%	16/24 = 67%

Verification precision exceeds generation recall in every arm. Bootstrap separates the paired gap for Claude (+52.5 points), GPT-5.6 Sol (+26.3) and Gemini (+23.3); Grok’s gap (+9.2) is directional. A clean-clone control for Grok bounds key leakage (McNemar $p=0.39$). Reasoning-effort tier per vendor arm was not recorded (gap #2).

11.3 Literature decomposition (copied, not re-derived)

Any system reporting top-1 = a and top- k = b implies recall $\geq b$ and conditional precision $\leq a/b$. Script: scripts/literature_decomposition.py.

Table 14: Selected literature decomposition rows (as in main-text Table 6).

system (<i>n</i>)	top-1	top- <i>k</i> (<i>k</i>)	recall	prec.
this work, solver alone (300)	39.3%	44.3% (3)	44.3%	88.7%
this work + forward-verify (194)	29.9%	33.5% (3)	33.5%	89.2%
NMR-Solver, literature (450)	52.9%	67.3% (10)	$\geq 67.3\%$	$\leq 78.6\%$
NMRAgent, literature (450)	61.6%	70.0% (10)	$\geq 70.0\%$	$\leq 88.0\%$
Espejo Morales, education (236)	80.9%	90.0% (5)	$\geq 90.0\%$	$\leq 89.9\%$
Espejo Morales, industrial (34)	20.6%	29.1% (5)	$\geq 29.1\%$	$\leq 70.9\%$
Alberts, 6–13 atoms, NIST IR (3,455)	63.8%	84.0% (10)	84.0%	75.9%
SpecX, scaffold split, simulated (99,439)	29.7%	50.6% (10)	50.6%	58.7%

Three groups changed only the data, and recall carried 68% (NMR-Solver simulated $\rightarrow$ real), 70% (SpecX random $\rightarrow$ scaffold) and 83% (Espejo education $\rightarrow$ industrial) of each collapse. The solver-alone precision column on $n=300$ is self-ranking (118/133); the forward-verify row is the instrumented $n=194$ slice (58/65).

12 Two-paper split and anonymity

IRSpectra-Bench (this ICLR paper) is the blind peak-list benchmark and the propose $\gg$ verify diagnosis. IRexp is a companion Scientific Data Data Descriptor (in preparation): 121,233 records; 43,060 structure-linked; 33,201 full IR+ ^1H + ^{13}C +structure quadruples. This SI cites that resource and does not re-present Methods / Data Records / licence-pool narrative.

Double-blind review copy: <https://anonymous.4open.science/r/peaklist-corpus-review-10C4/>. Named Hugging Face / GitHub hosting is restored at camera-ready. Do not put ilkhamfy/* URLs in the PDF. Fine-tunes must use data/train_no_bench.jsonl.gz so benchmark InChIKeys are withheld.

Listing 1 is the only IRexp-style JSON+molecule plate in the ICLR PDF. It is a pointer, not a second paper.

13 Reproducibility checklist

ICLR-style checklist, filled honestly. “N/A” is used only when the item does not apply; gaps are named.

1. Code / data / frozen deposits. Frozen questions / predictions for $n=194$ and +106 live in spectro-agent. This postcard repo is the ICLR manuscript. Camera-ready will name the archival host.
2. Headline regeneration. python scripts/score_pooled.py regenerates 118/300 and 133/300 when the withheld +106 key is restored locally.
3. Locked-slice fverify. python scripts/forward_verify_all.py regenerates 58/7/129 on $n=194$ only (SI protocol-slice; not the main-text $n=500$ generation wall).
4. expand-500 score2. Restore /tmp/blind/_key/benchmark_expand_500.answers2.jsonl.withheld, then benchmark_v2.py score2 -outdir data/benchmark_expand_500. Re-withhold the key. Confirm git ls-files data/benchmark_expand_500/answers2.jsonl is empty.
5. Do not commit answers2.jsonl for +106 or expand-500.
6. Do not invent CIs for expand-500; -ci was not run.
7. expand-500 fverify score is already in NIGHT_POOL / fverify STATUS (103/230 verify vs 129/230 self). Do not invent CIs or a wall. Do not re-run official prep.
8. Double-blind. Companion bib is Anonymous; dataset URL in the PDF is only the anonymous.4open review copy.

9. Fine-tunes must use data/train_no_bench.jsonl.gz.
10. IRexp construction, licences, and transcription audit belong in the Scientific Data companion — not in this SI beyond Listing 1.
11. Inference is not bit-exact. No pinned snapshot, temperature, or seed (Limitations (i)). Distributional agreement of frozen deposits is the intended claim.
12. Uncaptured artefacts (cannot be recovered): headline Claude wrapper text; dated model snapshots; decoding parameters; main-round sampler seeds; closed-book audit transcripts (docs/SUBMISSION.md gaps #1–#5).
13. Deferred arms. Expert-chemist audit not run. Leave-one-modality-out abandoned. Parser extraction-recall audit deferred to Sci. Data. SpectraLLM / IR-Agent / Spectro / NMIRacle / Alberts / CASE not scored on this bench.
14. Figure rebuild limits. Canonical Fig. 1 / fig_wall_diagnostic is the $n=500$ generation wall (227/22/251), not rebuilt from $n=194$ diagnosis.json. expand-500 has an official score() but no pooled fverify diagnosis.json. Size / recency / miss-isomer / named-ring analyses were never written against the expansion roster.
15. This manuscript compile. python3 scripts/build_pdf.py (main); python3 scripts/build_si.py (this SI). Engine: tectonic 0.17.0 or pdfLaTeX+BibTeX / XeLaTeX+BibTeX.

13.1 What a reviewer can re-check without the withheld keys

- Every a/b ($c\%$) in the main text and this SI is internally consistent with the integers in docs/HEADLINE_N500_2026-09-20.md, docs/POOLED_HEADLINE_2026-09-16.md and docs/NIGHT_PC_2026-09-20.md.
- Fig. 1 caption, Table 2, Limitations (vi), and Appendix FAQ agree: wall = $n=500$ generation (227/22/251); headline = $n=500$ generation (227/500, 249/500). Old fverify 58/7/129 is this SI protocol-slice only.
- expand-500 all-230 rows carry “no CIs / thinking-tier / not a wall”. Official fverify is 103/230 verify vs 129/230 self (SI only; not a wall). Do not quote 243/519.

14 Reviewer FAQ (SI copy)

These answers restate main-text Limitations and Appendix B. They add no new metrics.

What does the wall figure show? Figure 1 / fig_wall_diagnostic are the $n=500$ generation decomposition: 227 exact top-1, 22 in pool not top-1, 251 never proposed in top-3 (249 recalled, 49.8%; $45.4\% = 49.8\% \times 91.2\%$). Instrumented fverify 58/7/129 is this SI protocol-slice, not that wall.

Is the expansion hidden in the pool? No. Expansion top-1 is 59% vs 28% on the locked 194. That gap is a result.

Where is forward-verify on the 106? Not run. Conditional 89% (58/65) is $n=194$ only.

Where is the all-230 expand-500 top-1? This SI / main appendix: 129/230 (56.1%) top-1, 138/230 recall; 200-cut 109/200. Thinking-tier arm; no CIs. Official fverify: 103/230 verify vs 129/230 self (§8). Headline is $n=500$ (Eq. 7), not 230 and not 519.

Consumer harness / no snapshot / no thinking tier? Correct and intentional for the scored headline. Scoring of frozen deposits is reproducible; inference is not bit-exact.

Why no SpectraLLM / IR-Agent / CASE SOTA table? Different objects. None is scored on this bench. The paper is a stage diagnosis, not a method race.

Is IRexp a second ICLR paper? No. IRexp is a companion resource pointer (Listing 1; §12).

Should the headline move to $n=524$ or quote 243/519? No. Eq. 7 uses $\mathcal{S}$ of 500 (194+all 106+200). 524 / 530 are sensitivity. 243/519 drops five +106 flags — do not quote it.

What this SI is not

A Data Descriptor. A pooled wall. An expand-500 wall. A recommendation to switch the ICLR headline to $n=524$ or 243/519. A source of invented CIs or new metrics.